

Teertha Mazumder

CSA II @ OCIO | Government of Newfoundland & Labrador

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PROFESSIONAL SUMMARY

Business and Technology Analyst with experience delivering enterprise IT solutions within the Government of Newfoundland and Labrador. Demonstrated expertise in leading requirements of elicitation, business and process analysis, and supporting structured project delivery within public sector environments. Skilled in translating complex business, operational, and technical requirements into secure, scalable, and practical solutions across infrastructure, applications, and data platforms. Experienced in operating within governance frameworks, aligning solutions with enterprise standards, and collaborating effectively across technical teams, business clients, and leadership stakeholders. Holds a Master of Applied Science in Software Engineering and brings strong analytical judgment, structured documentation expertise, and the ability to bridge strategic business objectives with technical implementation.

PROFESSIONAL EXPERIENCE

Computer Systems Analyst II

OCIO- Government of Newfoundland and Labrador

July 2025 – Present

- Support enterprise infrastructure and application environments across development, testing, and production within OCIO governance frameworks.
- Business & Project Delivery Contributions
- Collaborate with business clients, project managers, and technical teams to clarify and document business, functional, and non-functional requirements (availability, performance, recovery, security).
- Conduct impact analysis for infrastructure and application changes, including OS upgrades, virtualization changes, and service deployments.
- Translate business and operational needs into structured technical implementation plans across Linux (RHEL), IBM AIX, and VMware environments.
- Participate in solution design discussions to ensure requirements are practical, testable, and aligned with enterprise standards.
- Support lifecycle delivery activities including planning, implementation, validation, and transition to operations.
- Perform root cause analysis on complex system incidents; document findings and recommend corrective and preventive measures.

- Contribute to disaster recovery testing, security compliance validation, and business continuity planning.
- Coordinate with vendors regarding enterprise services (Apache, Tomcat, Oracle, MySQL, IBM Spectrum Protect).
- Produce structured technical and operational documentation suitable for governance and audit review.
- Provide on-call support and manage competing operational priorities within enterprise delivery timelines.

Technical Environment: RHEL, IBM AIX, VMware vSphere, IBM Power Systems (HMC/LPAR), Apache, Tomcat, Oracle, MySQL, IBM Spectrum Protect.

Data Analyst / Engineer (Contract)

PROSHIKA – A Center for Human Development | Dhaka, Bangladesh

September 2020 – July 2023

Led business analysis and reporting system improvement initiatives supporting organizational planning and operational decision-making.

- Conducted stakeholder consultations to gather and refine reporting and system enhancement requirements.
- Defined business rules, validation logic, and data governance standards for reporting processes.
- Developed current-state and future-state workflow models to improve reporting efficiency and accuracy.
- Performed impact analysis to assess reporting changes on downstream operational processes.
- Designed and implemented automated data pipelines using Python, improving data integrity, and reducing manual effort.
- Developed executive dashboards using Power BI to support leadership-level decision making.
- Coordinated cross-functional validation sessions and supported user acceptance validation.
- Documented system logic, data definitions, and workflow processes to support governance and continuity.

Technical Stack: Python, SQL, Power BI, Data Modeling, ETL Processes

ACADEMIC & ENTERPRISE PROJECT/RESEARCH EXPERIENCE

Electronic Design Automation Using LLM (2024) | Memorial University of Newfoundland

Conducted structured analysis to identify inefficiencies in electronic design automation workflows and evaluate AI-driven optimization approaches.

- Defined functional and non-functional requirements for AI-assisted circuit design automation.

- Developed proof-of-concept models using Python, LLaMA-based LLM frameworks, and data preprocessing pipelines.
- Applied prompt engineering techniques and structured output validation logic.
- Used Python libraries for data handling and model evaluation.
- Designed validation metrics to measure model accuracy, output consistency, and performance improvements.
- Documented architecture, evaluation methodology, and risk considerations.

Technical Stack: Python, LLM (LLaMA), NLP techniques, data preprocessing, evaluation of metric design.

Demonstrated: Requirements analysis, feasibility study, AI prototype implementation, validation modeling.

Skier Management Web Application (2024) | Memorial University of Newfoundland

Developed a web-based athlete tracking and performance management system.

- Conducted requirements for elicitation sessions to define system features and reporting needs.
- Modeled workflows using use-case diagrams and relational data mapping.
- Designed backend architecture using Node.js (Express framework).
- Developed frontend using React.js with REST API integration.
- Designed relational database schema (SQL-based).
- Implemented authentication and data validation mechanisms.
- Conducted functional testing and debugging cycles.
- Produced user documentation and deployment guidelines.

Technical Stack: Node.js, Express, React.js, REST APIs, SQL database, JavaScript, Git

Demonstrated: Full-stack development, workflow modeling, database design, functional validation

Cost-Optimized NAS Infrastructure Solution (2020)

Designed and evaluated a cost-efficient network-attached storage architecture.

- Performed infrastructure requirement analysis and feasibility assessment.
- Designed storage architecture using a Linux-based server environment.
- Configured file systems and storage allocation policies.
- Conducted performance benchmarking and cost analysis.
- Developed deployment and configuration documentation.
- Implemented scripting (Bash) for automation and monitoring.

Technical Stack: Linux Server, Bash scripting, Storage configuration, Network file systems, Cost analysis modeling

Demonstrated: Infrastructure planning, performance validation, and technical feasibility analysis.

Online Result Management System (2019) | Daffodil International University

Designed and implemented an academic result publication platform.

- Gathered functional requirements from academic administrators.
- Defined grading business rules and role-based access control policies.
- Designed normalized relational database schema.
- Developed backend logic using PHP.
- Implemented frontend using HTML/CSS.
- Integrated MySQL database with validation logic.
- Conducted system testing to ensure accuracy and integrity of published results.

Technical Stack: PHP, MySQL, HTML/CSS, SQL, database normalization techniques

Demonstrated: Business rule implementation, data validation, secure access control design.

Additional Technical Projects

1. Knowledge Graph Modeling

- Extracted structured entities from Twitter microdata.
 - Built relationship mapping using graph modeling concepts.
 - Implemented data parsing and transformation using Python and JSON processing.
- Tech: Python, JSON parsing, data modeling concepts

2. Sentiment Analysis

- implemented a supervised machine learning model.
 - Preprocessed text datasets and applied classification algorithms.
 - Evaluated model performance using accuracy metrics.
- Tech: Python, Scikit-learn, NLP techniques

3. Voice-Activated Automated Car

- Designed embedded system logic for obstacle detection.
 - Integrated voice recognition input with a microcontroller-based control system.
- Tech: Microcontroller logic, sensor integration, embedded programming

4. Data Link Layer Simulation

- Implemented transmission protocol simulation in Java.
 - Modeled data frames, error detection, and retransmission logic.
- Tech: Java, network protocol simulation

5. E-Commerce Android Application

- Designed SQLite database schema for product and order management.
- Implemented Android frontend with transactional logic and validation.

Tech: Android (Java), SQLite, database schema design

EDUCATION

- Master of Applied Science (Software Engineering) | Memorial University of Newfoundland
- Bachelor of Science (Computer Science & Engineering) | Daffodil International University

TECHNICAL & ANALYTICAL SKILLS

- Operating Systems: RHEL, Ubuntu, IBM AIX, Windows, macOS
- Infrastructure & Virtualization: VMware vSphere, IBM Power Systems (HMC/LPAR)
- Programming & Prototyping: Python, JavaScript, PHP, Bash, Node.js, React, Flutter
- Data & Analytics: Power BI, Tableau, Excel, Pandas, MySQL, PostgreSQL, SQLite
- DevOps & Automation: Docker, Kubernetes, Ansible
- Cloud Platforms: AWS, Azure, GCP, Terraform, CloudFormation
- Machine Learning: Scikit-learn, TensorFlow, LLM Development

CERTIFICATIONS & TRAINING

- Red Hat Certified System Administrator (RHCSA)
- Linux System Engineering & Security
- Ansible Configuration Management
- Linux Networking & Patch Management
- Data Analyst & Development Certificate
- Excel Essential Training (NASBA)
- Emergency First Aid (CPR-B & CPR-C)

ADDITIONAL PROFESSIONAL EXPERIENCE

- Homecare Support Worker | Provincial Homecare
- Food Counter Attendant | Tim Hortons
- Cashier | Michaels

Developed strong real-world operational awareness and customer-focused communication skills in high-volume service environments.

- Worked directly with diverse clients and stakeholders, adapting communication styles to different needs and expectations.
- Managed competing priorities in fast-paced, deadline-driven environments.
- Developed practical understanding of frontline service delivery, operational workflows, and customer experience challenges.
- Strengthened conflict resolution, time management, and problem-solving abilities.
- Gained insight into how business processes function at the operational level and how service delivery impacts end users.